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Research on user trust evaluation of social e-commerce platform based on artificial neural network

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ABSTRACT

Due to the problems of poor evaluation accuracy and low evaluation efficiency existing in traditional social e-commerce platform user trust evaluation methods, a social e-commerce platform user trust evaluation method based on artificial neural network is proposed. According to the business factors, environmental factors and social factors, the evaluation system of user trust degree of social e-commerce platform is constructed, so as to build the objective function of user trust degree evaluation index weight calculation of social e-commerce platform. The objective function is optimized by using artificial neural network to obtain the evaluation index weight, and the evaluation model is constructed combined with the calculation results, complete the user trust evaluation of social e-commerce platform. Simulation results show that the proposed method has higher accuracy and shorter evaluation time.

1. Introduction

With the continuous development of science and technology and the continuous progress of computer technology, offline social activities have turned into online social activities, so social networks are developing rapidly. Social network is not limited by geography and time and space, and has the advantages of information sharing, timely communication and accurate positioning. Consumers chat and shop through social e-commerce platform, which provides great convenience for life [1]. Social e-commerce is a new e-commerce model based on e-commerce. It uses social media and websites as carriers to sell products. Social e-commerce platform with a unique sales model makes people gradually evolve from offline shopping to online shopping, making the shopping experience better. Shopping speed is more convenient and fast, so that users can buy products that are difficult to buy in the past [2]. However, in the process of shopping, social network users will worry about the disclosure of personal privacy information, thus reducing the willingness to buy, thus reducing the transaction rate. Therefore, trust has become the most direct problem between social network users and e-commerce platforms. The development of social e-commerce is positively correlated with the trust degree of users. Higher trust degree of users will further enhance the economic volume of social e-commerce, while lower trust degree of users will make users stay away from the social e-commerce mode, resulting in a downward trend of social e-commerce economy. In order to improve the service quality of social e-commerce platform, it has become the primary problem to study the evaluation method of user trust of social e-commerce platform [3].

Reference [4] proposes a social e-commerce platform user trust evaluation method based on BP learning. First, it combines web crawlers and association rule mining methods to collect social e-commerce platform data, and builds a social e-commerce platform

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user trust evaluation system based on the collection results. Obtain the first-level indicators of social e-commerce platform user trust evaluation, and input the indicators into the BP neural network for learning, obtain fuzzy constraint parameters, and on this basis, use machine learning methods to build social e-commerce platform user trust evaluation model. Reference [5] proposes a social e-commerce platform user trust evaluation method based on BP neural network, and establishes an evaluation system based on three first-level indicators of social e-commerce platform user trust evaluation, including honesty, goodwill, and ability, to obtain evaluation second-level indicators. Through the BP neural network, the user trust of social e-commerce platform is evaluated, and the evaluation result is obtained. Although the above two methods both achieve the evaluation of user trust in social e-commerce platforms, the evaluation results are quite different from the actual test results, resulting in low evaluation accuracy and unsatisfactory evaluation results. Reference [6] proposes a social e-commerce platform user trust evaluation method based on Internet of Things nodes. According to the influence of e-commerce platforms on social network user trust, obtain social e-commerce platform user trust evaluation indicators, and according to the Internet of Things node trust degree, calculate the weight of the evaluation index, and build a social e-commerce platform user trust evaluation model according to the obtained weight coefficient. Reference [7] proposed a method for evaluating user trust in social e-commerce platforms based on fuzzy mathematics, and obtained evaluation factors through the established evaluation system of user trust in social e-commerce platforms, and based on the influence of different evaluation factors on user trust in social e-commerce platforms, calculate the weights of evaluation factors, and use fuzzy mathematics methods to evaluate the trust of users of social e-commerce platforms. However, the above two methods take a long time to evaluate the trust of users of social e-commerce platforms, resulting in low evaluation efficiency.

In view of the problems of the above methods, this paper uses the advantages of artificial neural network method, such as fast, effective and accurate, to evaluate the user trust of social e-commerce platform, so as to improve the evaluation accuracy, solve the problems of traditional methods, and provide security for social e-commerce platform. The specific research route of this method is as follows:

(1) According to the business factors, environmental factors and social factors, the user trust evaluation system of social e-commerce platform is constructed, and the evaluation indexes are obtained, including 12, including business reputation, business service, interest, usefulness, usability, website quality, privacy and security;.

(2) Through the artificial neural network structure, according to the obtained user trust evaluation index of the social e-commerce platform, the target function is calculated by the weight of the user trust evaluation index;.

(3) The objective function is optimized by artificial neural network;.

(4) According to the obtained optimal solution, the weight of user trust evaluation index of social e-commerce platform is calculated, and the user trust evaluation model of social e-commerce platform is constructed.

(5) The evaluation accuracy and time of different methods are compared by experiments.

2. User trust evaluation of social e-commerce platform

2.1. Construction of user trust evaluation system on social e-commerce platform

Social e-commerce platform user trust refers to the degree of trust that social network users have in social e-commerce platforms, it is a quantitative value. The trust of social network users in social e-commerce platforms is not just a simple issue of trust. It is a question of how much trust can be achieved, so the degree of trust of social e-commerce platform users needs to be measured by trust [8]. For social e-commerce platforms, by establishing a social e-commerce platform user trust evaluation system, the influencing factors of social network users' trust in the social e-commerce platform are obtained, that is, social e-commerce platform user trust evaluation indicators, according to the evaluation the indicator builds an evaluation model to improve the service quality of social commerce platforms, and can better provide references and recommendations for social network users to purchase goods [9]. The user trust evaluation indicators of social e-commerce platforms are as follows:

(1) Merchant factors

For the trust of social e-commerce platform users, merchant factors occupy a large proportion. Pre-sales and after-sales service attitudes will affect the trust of social network users in social e-commerce platforms. Therefore, merchants build awareness through reputation and services to enhance social networking. Internet users' trust in social e-commerce platforms [10]. The higher the reputation and service quality, the higher the social e-commerce platform user trust; the lower the reputation and service quality, the lower the social e-commerce platform user trust. Reputation and service quality are positively correlated with the social e-commerce platform user trust [11]. In summary, as an important and classic factor that affects the trust of users of social e-commerce platforms, the reputation and service of merchants are included in the trust evaluation model as two secondary indicators of merchant factors.

(2) Envirnmental factor

One factor that affects the trust of users of social e-commerce platforms is environmental factors. When social network users make purchases on social e-commerce platforms, they will consider the security, personalization, and practicality of the platform. Social network users will Interaction in e-commerce platforms, looking for friends with the same interests, and sharing product information with friends, social network users experience the fun of shopping can enhance the trust in social e-commerce platforms [12].

(3) Socialization factor

For social e-commerce platform users' trust, one of the most important factors is socialization. When social network users make purchases on e-commerce platforms, friends will give corresponding opinions and discuss them in the community [20]. Internet users will be affected to a certain extent, thereby generating trust in social e-commerce platforms and enhancing the trust of users of social e-commerce platforms. Therefore, socialization factors are defined according to four indicators: social participation, social interaction, social recommendation and social identity [13].

Based on the above, this article constructs a social e-commerce platform user trust evaluation system from three aspects: merchants, online trading environment and social networks. If the social network users have a high degree of trust in the social e-commerce platform, it means that the social network users are very satisfied with the service quality of the social e-commerce platform, have a high sense of security on the social e-commerce platform, and have a high degree of confidence in the products on the social e-commerce platform. I am satisfied with the product information recommended by the social e-commerce platform [14]. According to the user trust evaluation system of social e-commerce platforms, the evaluation indicators are obtained [15]. Then the user trust evaluation system of social e-commerce platform is shown in Table 1:

2.2. User trust evaluation of social e-commerce platform based on artificial neural network

2.2.1. Artificial neural network structure

The artificial neural network method is different from the traditional artificial intelligence technology. It is a computational model that has the advantages of intelligent learning, strong self-adaptation, and fast calculation solution optimization speed. It solves the defects of traditional artificial intelligence in unstructured information. The calculation time is shortened, and the optimal solution of the problem can be obtained in the shortest time. The artificial neural network is composed of countless neurons, which are connected to each other, have the ability to process information, and each neuron represents a different meaning [16]. The artificial neural network structure is shown in Fig. 1:

As shown in Fig. 1, the artificial neural network can complete any n-dimensional to m-dimensional mapping. The artificial neural network structure includes an input layer, an output layer and a hidden layer. The neurons in the input layer represent the input unit, the neurons in the output layer represent the output unit, and the neurons in the hidden layer represent the hidden unit, which mainly processes and transmits information [17].

The artificial neural network obtains external data through the input layer, and sets the input vector of the artificial neural network input layer:

$$I = (i_1, \ldots, i_n)^T \tag{1}$$

Use the hidden layer to transfer the processed information, and set the hidden layer output vector:

$$C = (c_1, \ldots, c_m)^T \tag{2}$$

Set the output vector of the output layer:

$$O = (o_1, \ldots, o_l)^T \tag{3}$$

Through the interconnection between neurons, the neuron weight vectors w_j^1 and w_k^2 are obtained, and the external information is processed, then the expression of the weight matrix between the input layer and the hidden layer is:

$$W^1 = \left(w_1^1, \ldots, w_j^1, \ldots, w_m^1\right) \tag{4}$$

Neurons will transmit the information they produce to neighboring neurons, and then use the artificial neural network composed of neurons to process the information and output it. Then the expression of the weight matrix from the hidden layer to the output layer in the artificial neural network is:

Table 1
User trust evaluation system of social e-commerce platform.

First level indicator	Secondary indicators
Merchant factors	Business reputation x_{11}
	Merchant service x_{12}
Envirnmental factor	Fun x_{21}
	Usefulness x_{22}
	Ease of use x_{23}
	Website quality x_{24}
	Privacy and security x_{25}
	System guarantee x_{26}
Socialization factor	Social participation x_{31}
	Social interaction x_{32}
	Social recommendation x_{33}
	Social identity x_{34}

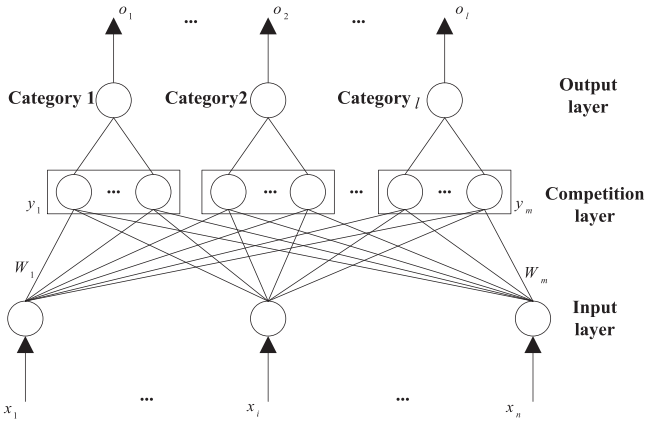


Fig. 1. Artificial neural network structure.

$$W^2 = \left(w_1^2, \ldots, w_k^2, \ldots, w_l^2\right) \tag{5}$$

In the formula, k represents the number of neurons [18].

According to the above artificial neural network structure, in the evaluation of user trust in social e-commerce platforms, business factors, environmental factors, and social factors are used as input data sets, such as business reputation, business services, interest, usefulness, ease of use, and website twelve social e-commerce platform user trust evaluation indicators including quality, privacy and security are used as output data sets.

Input the evaluation index $x_1, x_2, \ldots, x_i, x_{no}$ into the input layer, set the neurons in the input layer to i, the neurons in the hidden layer to j, the threshold of this layer is θ_j, the activation function of this layer is $g(\bullet)$, then the output of this layer for:

$$h_j = g\left(\sum_{i=1}^{no} w_{ij} x_i + \theta_j\right) \tag{6}$$

In the formula, w_{ij} represents the weight generated by the interconnection between i and j [19].

Set the neuron in the output layer as o, the threshold value of this layer is θ_o, and the activation function of this layer is $f(\bullet)$, then the output of the only neuron in the output layer, that is, the evaluated social e-commerce platform user trust:

$$y = f\left(\sum_{j=1}^{nh} w_{jo} h_j + \theta_o\right) \tag{7}$$

In the formula, w_{jo} is the weight generated by the interconnection between j and o [20].

2.2.2. *User trust evaluation of social e-commerce platform*

Build the objective function of user trust evaluation index weight calculation of social e-commerce platform, use artificial neural network to optimize the objective function, obtain the evaluation index weight, combined with the calculation results to build an evaluation model, complete the evaluation of user trust of social e-commerce platform.

Set X to represent the number of samples in the output data set of the social e-commerce platform user trust evaluation index. Through the establishment of a regression model, all the index features are combined to extract M types of feature sets, and form a feature matrix $X_1, X_2, \ldots, X_M$. Define $y = [y_1, y_2, \ldots, y_N]^T$ as the memorable trust vector of the data set, and y_i as the trust value of the i-th index. Therefore, the adaptive correlation matrix δ is introduced to obtain the objective function of the user trust evaluation of the social e-commerce platform:

$$G(w_i, H_i) = \left\| \sum_{i=1}^{M} x_i^T w_i - y \right\|^2 \tag{8}$$

In the formula, w_i is the regression coefficient.

The above objective function can be optimized through the artificial neural network method. First, fix w_i, and use the artificial neural network to optimize the incidence matrix δ. The expression is:

$$\frac{\partial L}{\partial \delta} = 2F_{(i)}{}^T E_{(i)} + 2F_{(i)}\delta \tag{9}$$

In the formula, $E_{(i)}$ and $F_{(i)}$ represent the ith column vector of the custom matrix E and F respectively.

According to the obtained optimal solution of the objective function, calculate the weight of the user trust evaluation index of the social e-commerce platform:

$$Q(\alpha,\beta)=\frac{1}{M}\sum_{i=1}^{M}(\alpha\delta+X_M\beta) \tag{10}$$

In the formula, α,β is the weight coefficient, which is used to measure the degree of influence of the users of the social e-commerce platform on the various indicators in the evaluation system. According to the above formula, it can be seen that the evaluation index will have an impact on the trust of users of social e-commerce platforms. Therefore, combined with the weights of evaluation indexes, a user trust evaluation model of social e-commerce platforms is constructed.

$$h(\vartheta)=\frac{y+\sqrt{1-k}^{2}}{1+\lambda^{-\mu(\vartheta-k)}} \tag{11}$$

In the formula: ϑ represents the standard value of evaluation ability; y represents the logical relationship; λ represents the evaluation vector; μ represents the evaluation coefficient.

3. Simulation experiment analysis

In order to verify the performance of the artificial neural network-based social e-commerce platform user trust evaluation method proposed in this paper in practical applications, a simulation experiment analysis was carried out. The simulation experiment environment is shown in Table 2.

According to the above experimental background, the user trust evaluation method of social e-commerce platform based on artificial neural network, the evaluation method based on BP learning and the evaluation method based on fuzzy mathematics proposed in this paper are used to evaluate the user trust of social e-commerce platform, and the three methods of social e-commerce platform user trust evaluation results are compared with the actual test results, and the comparison results are shown in Fig. 2.

According to Fig. 2, the user trust evaluation results of social e-commerce platform users using the artificial neural network-based social e-commerce platform user trust evaluation method proposed in this paper are consistent with the actual test results, but the evaluation method based on BP learning and the evaluation method based on fuzzy. The results of mathematical evaluation methods for social e-commerce platform user trust evaluation results are quite different from the actual test results, which shows that the social e-commerce platform user trust evaluation method based on artificial neural network is proposed in this paper, the result is ideal and the evaluation accuracy is high.

In order to further verify the effectiveness of the method in this paper, the user trust evaluation method of social e-commerce platform based on artificial neural network, the evaluation method based on BP learning and the evaluation method based on fuzzy mathematics proposed in this paper are used to increase the trust of users of social e-commerce platform. The evaluation time is compared and analyzed, and the comparison result is shown in Fig. 3.

According to Fig. 3, the evaluation time of the user trust degree of the social e-commerce platform based on the artificial neural network user trust evaluation method of the social e-commerce platform proposed in this paper is within 10 s. The evaluation method based on BP learning and the evaluation method based on fuzzy mathematics are carried out. The user trust evaluation time of social e-commerce platform users is within 35 s and 25 s, respectively, which shows that the user trust evaluation time of social e-commerce platform users based on the artificial neural network based on the artificial neural network evaluation method is longer than that based on BP learning. The evaluation method and the evaluation method based on fuzzy mathematics of social e-commerce platform user trust evaluation time is short, and the evaluation efficiency of the method in this paper is high.

4. Conclusion

Due to the low accuracy and long evaluation time of traditional methods for social e-commerce platform user trust evaluation, the evaluation effect is poor and the evaluation efficiency is reduced. Therefore, this paper proposes a social e-commerce platform user trust evaluation method based on artificial neural network. Since business, environment and socialization are the most influential factors to the user trust of social e-commerce platform, this paper constructs a user trust evaluation system of social e-commerce platform based on these three factors, and obtains 12 user trust evaluation indicators of social e-commerce platform, including business reputation, business service, interest, usefulness, ease of use, website quality, privacy and security. On this basis, the objective function of user trust evaluation index weight calculation of social e-commerce platform is constructed, and the objective function is optimized by using artificial neural network to obtain the evaluation index weight. Combined with the calculation results, the evaluation model is

Table 2
Simulation experiment environment.

Serial number	Name	Specific requirements
1	Operating system	Windows 7
2	JDK environment	Java SE Development KET version
3	Development tools	Eclipse 3.7.1 +Android 4.0.3
4	Development language	Java XML
5	Database management software	SQLite 3
6	Operating platform	Windows and Linun versions

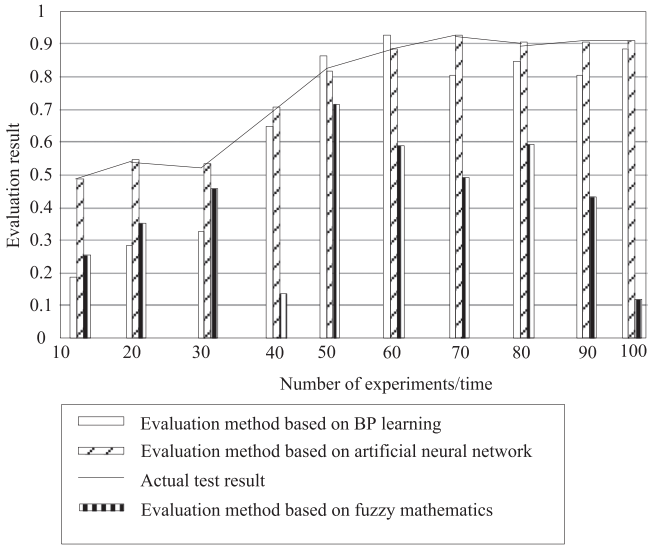


Fig. 2. Comparison results of the evaluation accuracy of the three methods.

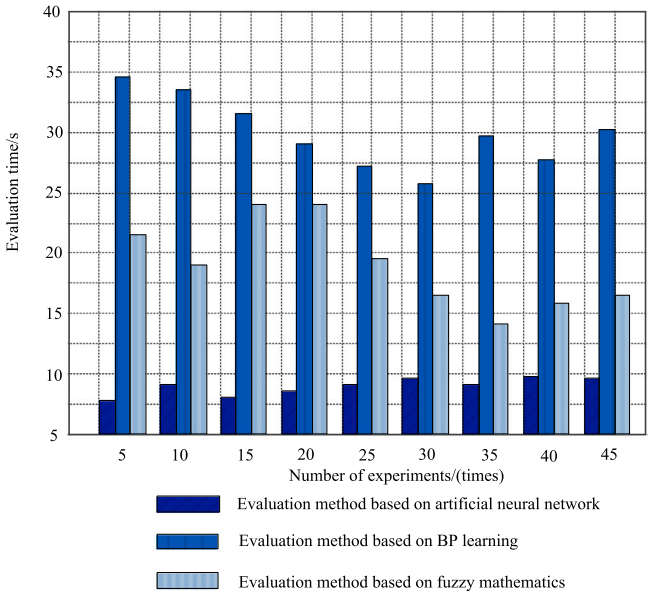


Fig. 3. Comparison results of the evaluation time of the three methods.

constructed to complete the user trust evaluation of social e-commerce platform. The simulation results show that the user trust evaluation method of social e-commerce platform based on artificial neural network is effective and efficient.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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